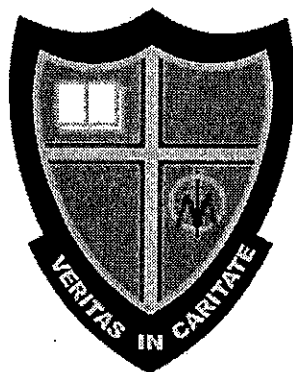


ST BENEDICT'S COLLEGE



MATHEMATICS
GRADE 11/12

STATISTICS CAMP

PAPER 2

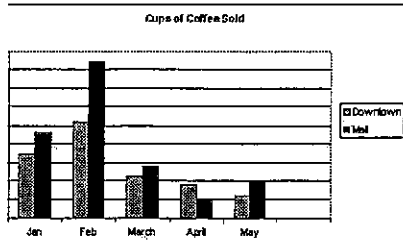
Grade 12 mark allocation: 20 ± 3 MARKS

DATA HANDLING

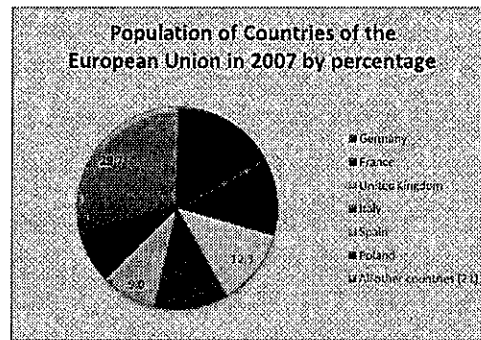
From previous grades:

Representing ungrouped data graphically:

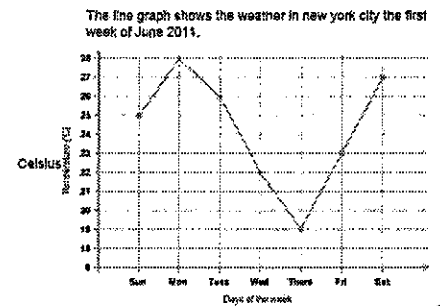
Bar Graph



Pie chart

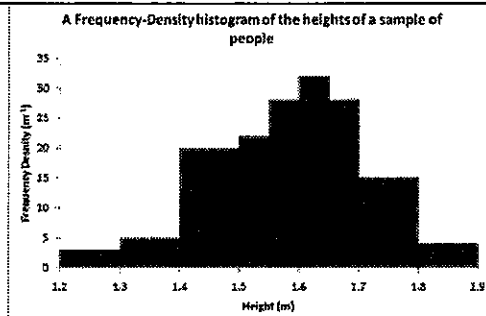


Broken line graphs

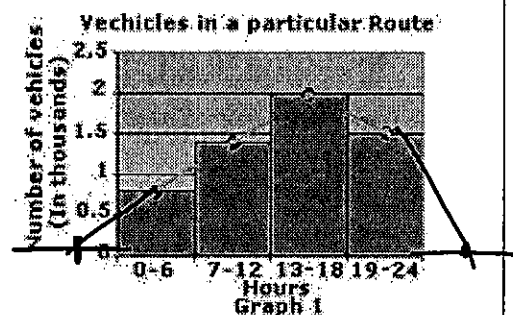


Representing grouped data graphically:

Histogram



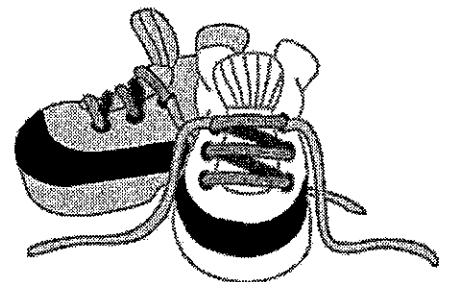
Frequency polygon



Measures of Central Tendency

Ungrouped data:

Mean	List of shoe sizes: 7,9,12,9,8,6,9,12,13,17	
Most commonly used measure of central tendency $\bar{x} = \frac{\sum x_i}{n}$ The mean is distorted by outliers		
Median		
Middlemost score (odd number) or average of the two middle score (even number)		
Mode		
The most frequent score Can have more than one mode		



Grouped data:

Estimate of the mean:

- Calculate the midpoint of each class
- Multiply each midpoint by the frequency for that interval
- Add up and divide by the total number of scores

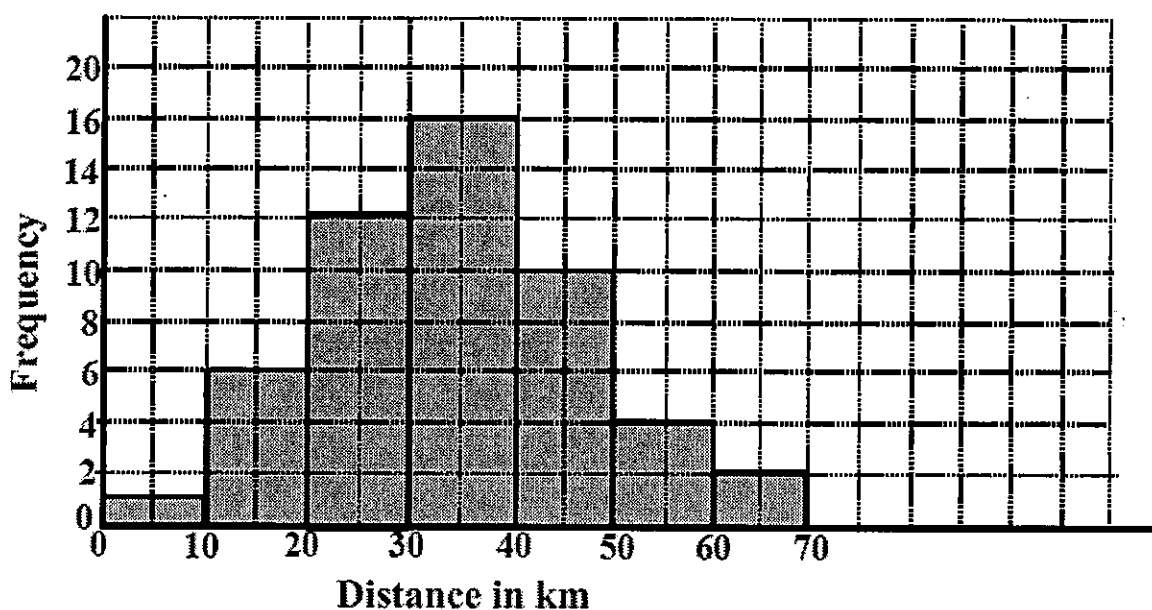
The modal class:

- This is the interval in which the data occurs most frequently

The median:

- The best way to calculate the median is by drawing a cumulative frequency curve (Ogive)
- A way of representing grouped data
- Never goes down and should form an S-shape
- Can also be used to estimate median, quartiles and percentiles

The following histogram represents the distance run by marathon runners.



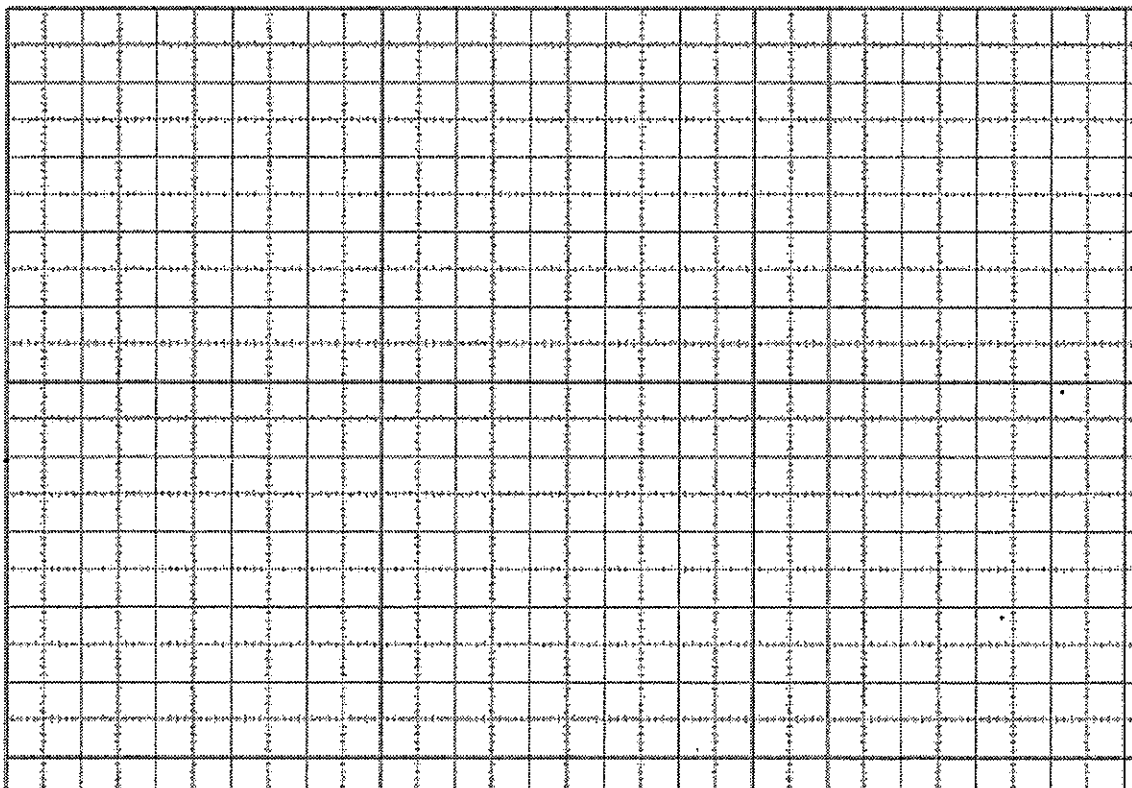
a) Complete the table below:

Class interval	Frequency
$0 < x \leq 10$	
$10 < x \leq 20$	
$20 < x \leq 30$	
$30 < x \leq 40$	
$40 < x \leq 50$	
$50 < x \leq 60$	
$60 \leq t \leq 70$	

(b) Calculate the estimated mean.

c) What is the modal class?

d) Construct a cumulative frequency curve and determine the median and the inter-quartile range.



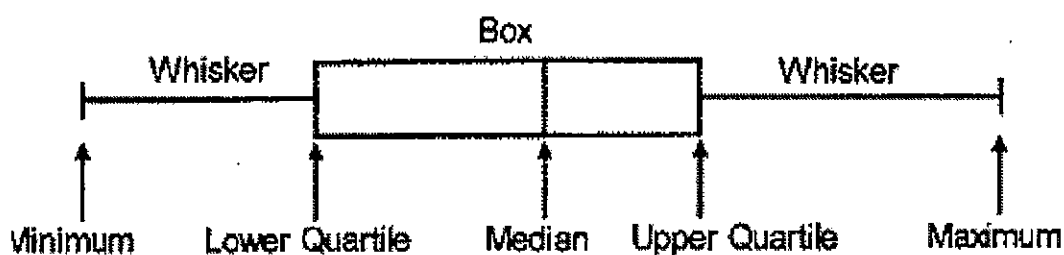
Measures of Dispersion: (Spread of Data)

Range	
Quartiles	
Deciles	
Percentiles	
Inter-quartile Range	
Semi- Interquartile Range	

Five Number Summary:

- Minimum value: Smallest value
- Lower quartile (Q_1): Median of the lower half of the data
Position of $Q_1 = \frac{n+1}{4}$
- Median: Position of $Q_2 = \frac{n+1}{2}$
- Upper quartile (Q_3): Median of the upper half of the data
Position of $Q_3 = \frac{3(n+1)}{4}$
- Maximum: Highest value

The Box and Whisker plot: (Remember to always draw the box-whisker plot on a number line)



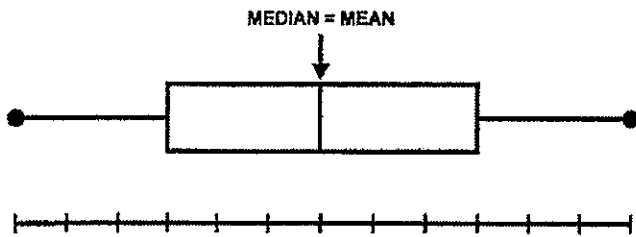
16 18 18 18 19 20 21 22 22 23 24 28 29

- ## Solutions

- | | |
|----------------------|------|
| Minimum: | 16 |
| Lower quartile (LQ): | 18 |
| Median: | 21 |
| Upper quartile (UQ): | 23,5 |
| Maximum: | 29 |

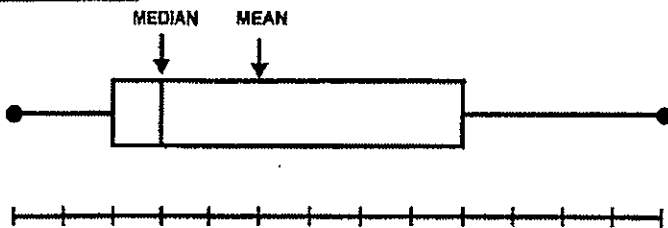
Distribution of Data:

Situation 1



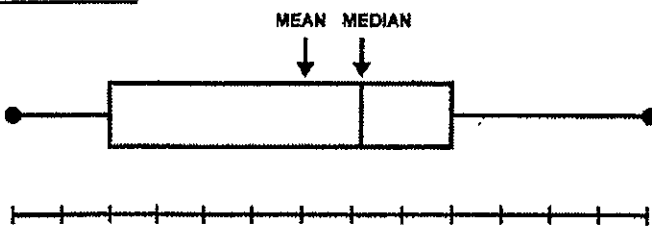
If the data to the left of the median balances with the data on the right, then the data is symmetrical about the median. The median will be very close to the mean and the data is said to be symmetrical about the mean. If the data is represented in a histogram, the mean, median and mode will be the same. The data is said to be normally distributed.

Situation 2



If there is a mass of data values predominantly on the left side of the distribution with fewer higher values on the right, the data is said to be positively skewed or skewed to the right. The mode will be closer to the left of the distribution. The median is to the left. In this case: $\text{mean} > \text{median}$

Situation 3



If there is a mass of data values predominantly on the right side of the distribution with fewer lower values on the left, the data is said to be negatively skewed or skewed to the left. The mode will be closer to the right of the distribution. The median is to the right. In this case: $\text{mean} < \text{median}$

Skewed Data and Outliers:

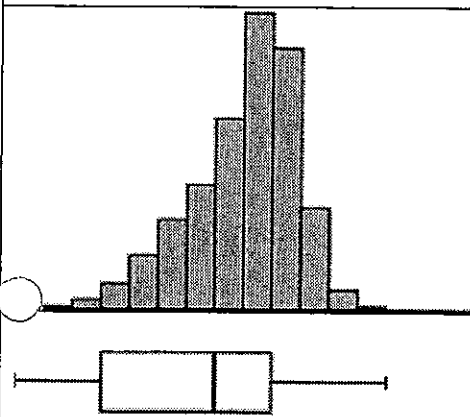
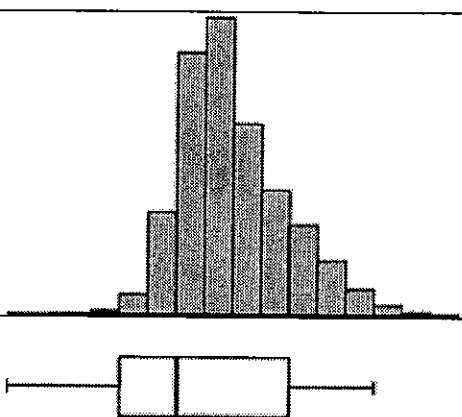
The mean is susceptible to the influence of outliers so if there are any outliers, the mean is not considered a good representation of the data.

If you have a normally distributed sample, the mean and median are both good measures of central tendency. (In perfectly symmetrical data the mean would equal the median)

If the data is skewed, the mean tends to be 'dragged' in the direction of the skewness. (In this case, the median is more likely to be a better representation of the data). Skewness exists if there are extreme scores or tail.

The more skewed the distribution, the greater the difference between the median and mean.

In most cases:

Negatively skewed (mean subtract median < 0)	Positively skewed (mean subtract median > 0)
Skewed to the <u>left</u> – the data is more spread out on the left (Longer tail on left = skewed to left)	Skewed to the <u>right</u> – the data is more spread out on the right. (Longer tail on right = skewed to the right)
mean < median < mode	mode < median < mean
	

Variance and Standard Deviation

- Spread of data
- Variance = average of squared differences of the mean
- Standard deviation = $\sqrt{\text{variance}}$

Standard deviation using a table (ungrouped data)

Example 1

The standard deviation (SD) can be determined by using the following formula: $SD = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$

x	$x - \bar{x}$	$(x - \bar{x})^2$
5	$5 - 7 = -2$	4
5	$5 - 7 = -2$	4
7	$7 - 7 = 0$	0
6	$6 - 7 = -1$	1
9	$9 - 7 = 2$	4
10	$10 - 7 = 3$	9
$\bar{x} = \frac{42}{6} = 7$	$\sum (x - \bar{x}) = 0$	$\sum (x - \bar{x})^2 = 22$

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n} = \frac{22}{7} = 3,6$$

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} = \sqrt{\frac{22}{7}} = 1,9 \text{ (rounded off to one decimal place)}$$

Standard deviation using a calculator (ungrouped data)

The standard deviation can be easily calculated by using a calculator.

CASIO fx-991ES PLUS

MODE

3 : STAT

1 : 1 - VAR

Enter the points:

Push = after each data point

AC

SHIFT

STAT (Press 1)

4: VAR

3 : σx

=

To clear CASIO calculator: MODE 1: COMP

CASIO fx-82ZAPLUS

MODE

2 : STAT

1 : 1 - VAR

Enter the data points:

Press = after each data point

AC

SHIFT

STAT (Press 1)

4: VAR

3 : $x \sigma n$

=

Standard deviation using a table (ungrouped data involving frequencies)

Example 2

The standard deviation (SD) can be determined by using the following formula: $SD = \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$

x	f	$f \times x$	$x - \bar{x}$	$(x - \bar{x})^2$	$f \times (x - \bar{x})^2$
4	3	12	$4 - 5,85 = -1,85$	3,4225	10,2675
5	7	35	$5 - 5,85 = -0,85$	0,7225	5,0575
7	10	70	$7 - 5,85 = 1,15$	1,3225	13,225
Totals	20	117	$\bar{x} = \frac{117}{20} = 5,85$		$\sum f(x - \bar{x})^2 = 28,55$

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n} = \frac{28,55}{20} = 1,4275$$

$$\text{Standard deviation} = \sqrt{\text{variance}} = \sqrt{\frac{28,55}{20}} = \sqrt{\frac{28,55}{20}} = 1,2 \text{ (rounded off to one decimal place)}$$

Standard deviation using a calculator (ungrouped data involving frequencies)

Whenever frequencies are given, the following calculator programmes apply:

CASIO fx-991ESPLUS

MODE

3 : STAT

1 : 1 - VAR

SHIFT SET UP

4:STAT (you will need to scroll down)

1: ON

Enter the marks

Enter the frequencies

AC

SHIFT STAT

4: VAR

3 : $\sigma x =$

CASIO fx-82ZAPLUS

MODE

2 : STAT

1 : 1 - VAR

SHIFT SET UP

3:STAT (you will need to scroll down)

1: ON

Enter the marks:

Enter the frequencies:

AC

SHIFT

STAT

4: VAR

3 : $x \sigma n$

=

If the data is grouped the middle value of the interval must be used as well as the frequency for the calculation as above.

EXERCISE 1:

1. The Comrades marathon is an ultramarathon of approximately 89 km which is run annually in Kwazulu-Natal between the cities of Durban and Pietermaritzburg. It is the world's largest and oldest ultramarathon race. The direction of the race alternates each year between the "up" run (87 km) starting from Durban and the "down" run (89 km) starting from Pietermaritzburg. The following table summarises the athletes who have achieved the most wins to date:

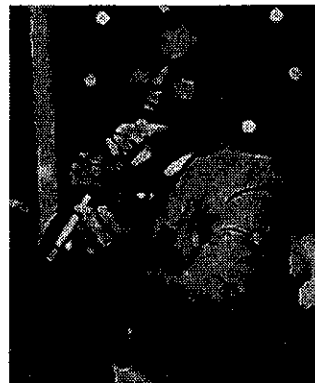
Athlete	No of wins	Athlete	No of wins
Alan Robb	4	Helen Lucre	3
Arthur Newton	5	Jack Mekler	5
Bruce Fordyce	9	Lettie van Zyl	3
Elena Nurgalieva	7	Maria Bak	2
Frith van der Merwe	3	Maureen Holland	4
Hardy Ballington	5	Wally Hayward	5

[Source: http://en.wikipedia.org/wiki/Comrades_Marathon]

- Calculate the mean for this data.
- What is the mode for the number of wins per athlete?
- Draw a box and whisker plot for this data.
- Comment on the distribution of the data in terms of skewness.
- Calculate the standard deviation for this data (one decimal place).
- How many athletes lie outside the first standard deviation interval?
- Why is Bruce Fordyce's number of wins an outlier? Give a mathematical reason to justify your answer.
- Draw a box and whisker plot for the data highlighting the outlier.

2. SA Idols is one of South Africa's biggest television shows. It has been a huge success and the winner of this singing competition for 2012 was twenty-five year old Khaya Mthethwa, who was the favourite throughout the competition. The statistics for the competition from the Idols Top 10 phase appear below with each contestant's number of votes represented by a percentage of the total.

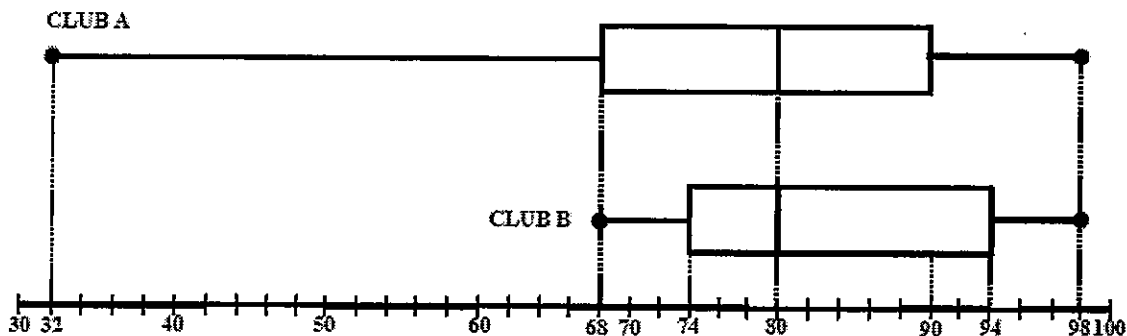
Nosipho	3,86%
Obakeng	3,00%
Monde	3,89%
Chloe	5,80%
Simphiwe	9,01%
Dominic	3,40%
Melissa	3,29%
Khaya	33,09%
Shekhinah	9,64%
Tshidi	25,01%



[Source: <http://idolssa.dstv.com>]

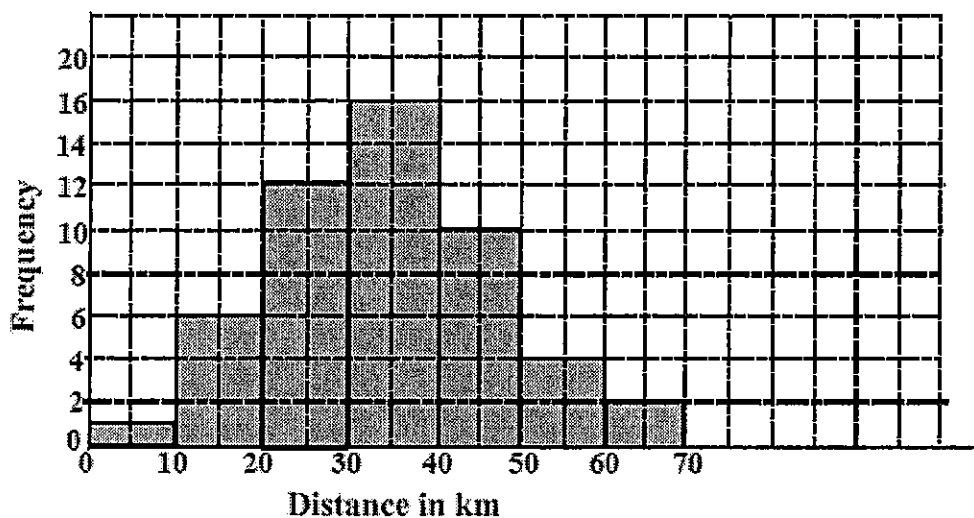
- Calculate the mean percentage for this data.
- Draw a box and whisker plot for this data.
- Comment on the skewness of the data.
- Which of the percentages are outliers?
- Redraw the box and whisker plot highlighting the outliers.
- Calculate the standard deviation for the data.
- Which percentages lie outside the first standard deviation interval?

3. The box and whisker plots below represent the distances run by athletes from two different running clubs over a period of one month.



- What features are the same for both clubs?
- It seems that there is no significant difference in the performance between the two clubs. Is this conclusion valid? Support your answer with reasons.
- Comment on the distribution of distances for Club A if the mean is 76.
- Determine whether the minimum or maximum values for Club A are outliers.
- Does Club B have any outliers? Explain.

4. The following histogram represents the distance run by a few Comrades Marathon runners who didn't complete the 89 km race.

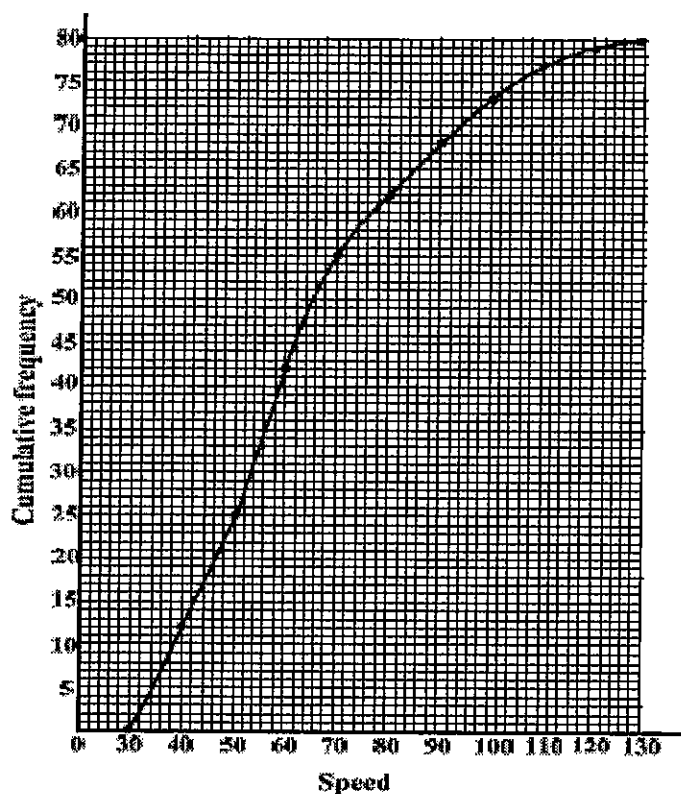


- (a) Redraw and complete the following table:

Class interval	Frequency	Cumulative frequency
$0 < x \leq 10$		
$10 < x \leq 20$		
$20 < x \leq 30$		
$30 < x \leq 40$		
$40 < x \leq 50$		
$50 < x \leq 60$		
$60 \leq x \leq 70$		

- (b) Calculate the estimated mean.
 (c) State the modal class.
 (d) Draw the cumulative frequency curve for this data.
 (e) Determine estimates for the quartiles.
 (f) Calculate the 60th percentile.
 (g) Draw a box and whisker diagram if the minimum distance is 10 km and the maximum distance is 70 km.
 (h) Comment on the distribution of the data in terms of skewness.

5. Residents in Knysna complained to the Police department that too many drivers were exceeding the speed limit of 60 km/h on a busy road near a school. The Police department decided to investigate the situation. The department recorded the speeds of drivers along this road over one week. The speed limit is 60 km/h. The following cumulative frequency curve represents the findings of the Department.

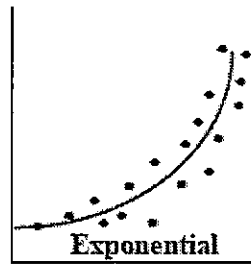
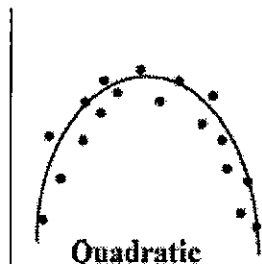
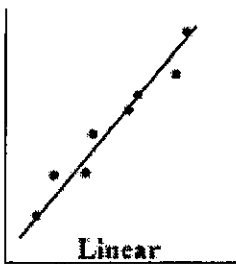


- (h) Calculate an estimated mean speed.
- (i) Calculate an estimated value for the standard deviation of this data.
(Hint: Use the midpoints of the class intervals as the x -values to enter in your calculator)

Scatter plots and line of best fit:

- Two sets of data where a relationship may be visible- bivariate data
- Relationship = correlation (r)
- Relationship can be linear, quadratic or exponential
- Linear relationships can be positive or negative and strong or weak (correlation coefficient)
- Line of best fit: $y = a + bx$

Scatterplots are useful diagrams that enable researchers to compare two sets of data to determine whether there is a relationship between them. For example, the relationship between a smoking habit and the chance of getting lung cancer can be investigated using a scatterplot diagram. The number of cars on a highway and the accident rate are two sets of data that might have a relationship or correlation. Plotting this data on a scatterplot diagram will show trends in the data. Data could follow a linear, quadratic or exponential trend.



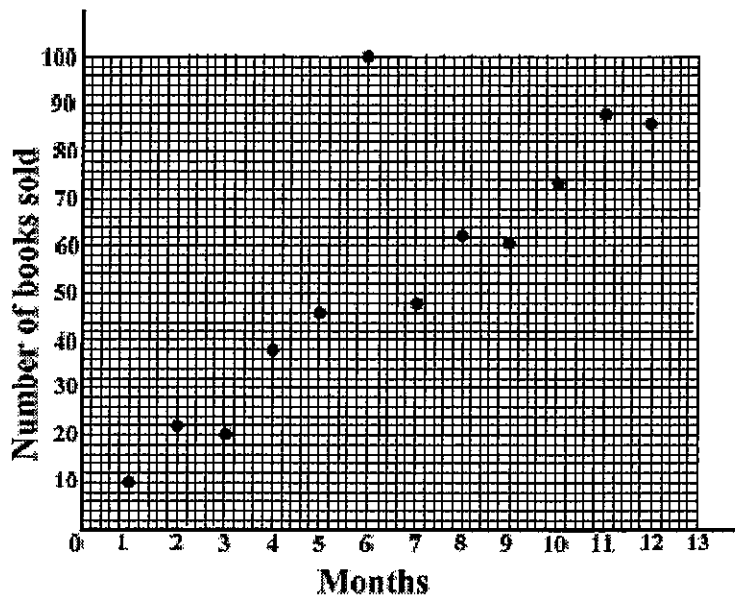
LINES OF BEST FIT

When data follows a linear trend, a line of best fit can be drawn on the scatterplot diagram. The following example will illustrate what the line of best fit is.

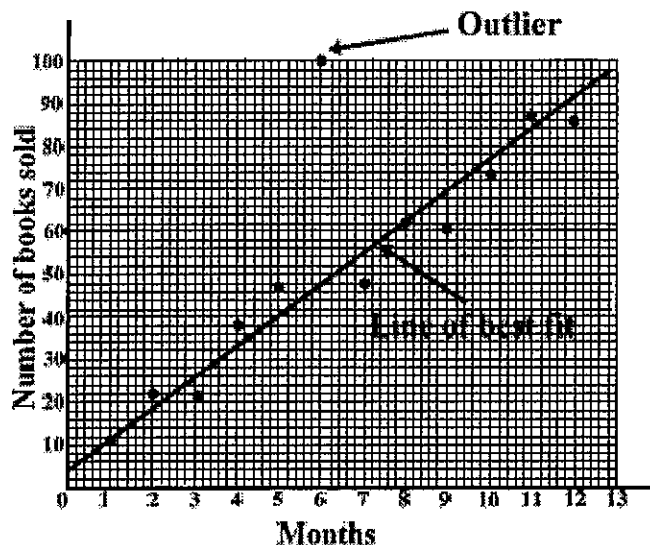
EXAMPLE 1

Consider the following scatterplot of information obtained by a publishing company that recorded the number of books sold over a period of 12 months. Investigate the relationship between the two sets of data: number of months and the number of books sold.

No. of months	1	2	3	4	5	6	7	8	9	10	11	12
No. of books sold	10	22	20	38	46	100	48	62	61	74	88	86



It is clear from the above graph that the points do not lie on a perfect straight line. The information on the graph does follow a linear trend. We can draw what is called a line of best fit, which will help to predict future values. In order to do this, draw a line through some of the points with the aim of having the same number of points above the line as below the line. A possible line of best fit will now be drawn onto the graph.



A line can be drawn through two of the points with five above and five below. A line of best fit will not represent the data perfectly, but it will give you an idea of the trend. Different people will probably draw slightly different lines of best fit, but the trend should be more or less the same in each case. One of the points (see diagram) is way out compared to the others. This point is called an outlier. Probably what happened during the sixth month was that the company had an unusually high sale of books. There might have been a conference where the company displayed and sold a lot of books. What is evident though, is that the number of books sold has a linear increase over the 12 months.

The line of best fit drawn in the diagram above is not really accurate since different people will have drawn different lines of best fit. What we will now demonstrate is a method to determine the actual equation of the line of best fit using the least squares method.

The least squares method involves determining the gradient and y-intercept of the line of best fit by using a calculator. The line of best fit is also called a regression line. The equation of this regression line (line of best fit) is given by $y = a + bx$ where a represents the y-intercept of the line and b the gradient or slope. A calculator can be used to determine the value of a and b . The outlier will need to be excluded as it is too far away from the line of best fit. * ONLY IF TOLD TO DO

A possible calculator programme is as follows:

MODE 2

PRESS: 2 : A + BX

ENTER DATA POINTS:

COLUMN (X)

1= 2= 3= 4= 5= 7= 8= 9= 10= 11= 12=

COLUMN (Y)

10 22 20 38 46 48 62 61 74 88 86

SO IN QUESTION.

THEN PRESS AC

THEN PRESS SHIFT 1

THEN PRESS 5: REG

THEN PRESS 1: A = TO GET THE VALUE OF a WHICH IS 5,315923567

THEN PRESS SHIFT 1

THEN PRESS 5: REG

THEN PRESS 2: B = TO GET THE VALUE OF b WHICH IS 6,896178344

THEN PRESS AC

THEN PRESS MODE 1 TO GET BACK TO NORMAL MODE

$y = 6,896178344x + 5,315923567$ is the equation of the line of best fit for this data, where the y-intercept is 5,315923567 and the gradient is 6,896178344.

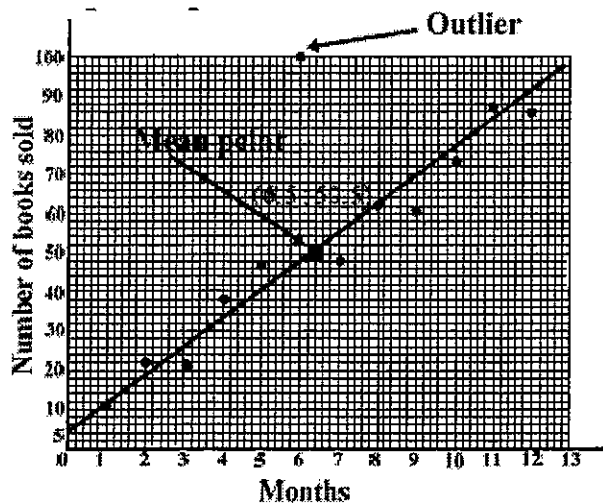
A very important principle is that the line of best fit passes through the mean point: $(\bar{x}; \bar{y})$

$$\bar{x} = \frac{1+2+3+4+5+7+8+9+10+11+12}{11} = 6,54545454 \approx 6,5$$

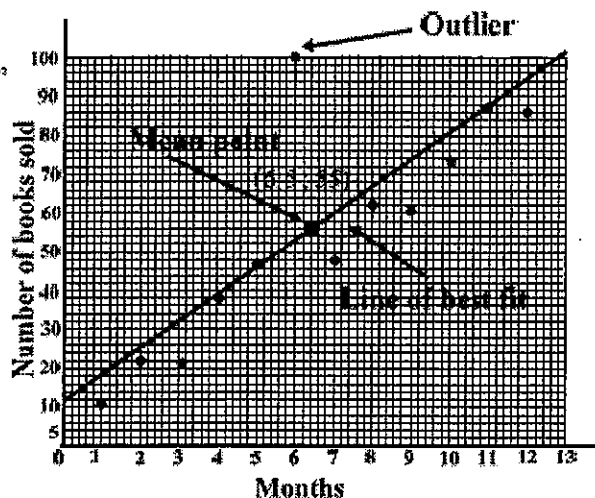
$$\bar{y} = \frac{10+22+20+38+46+48+62+61+74+88+86}{11} = 50,45454545 \approx 50,5$$

The line of best fit therefore passes through the point (6,5; 50,5)

In order to draw the line of best fit, plot the y-intercept and mean point and draw the line through these points.



Notice that if the outlier is included, the line of best fit will be "pulled upwards" and will not be accurate. The line will have an equation of $y = 6.70979021x + 10.96969697$ and the mean point will be (6.5, 55).



It is now possible to use the line of best fit to predict an estimate of the number of books that might be sold for the 13th month:

$$y = 6.896178344(13) + 5.315923567 \approx 94 \text{ books}$$

This process of estimating the number of books outside the given 12 months is called **extrapolation**. If the line of best fit is used to estimate the number of books inside the given 12 months, then this is called **interpolation**. For example, an estimate of the number of books sold after ten and a half months can be done as follows and is an example of interpolation:

$$y = 6.896178344(10.5) + 5.315923567 \approx 77 \text{ books}$$

Remember that this linear increase over the year might not necessarily continue in this manner. If the sales start dropping in the near future, the trend will no longer be a linear increase. It could change to a linear decrease (or even a quadratic trend) if the sales of books start dropping.

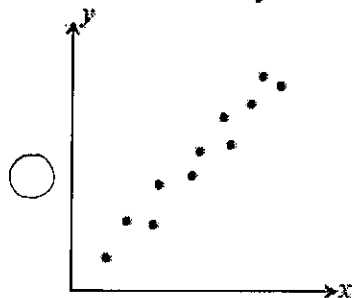
CORRELATION

The strength of the linear relationship between the two variables in a scatterplot depends on how close the data points are to the line of best fit. The closer the points are to this line, the stronger the relationship. If the points are further away from the line of best fit, the weaker the relationship. If the line of best fit slopes to the right and has a positive gradient, then the linear relationship is positive. If the line of best fit slopes to the left and has a negative gradient, then the linear relationship is negative.

The gradient of the line of best fit indicates whether the association is positive or negative. However, it will not indicate the strength of the association. The correlation coefficient (r) is a value that gives an indication of the strength of the association. It can be calculated using a calculator.

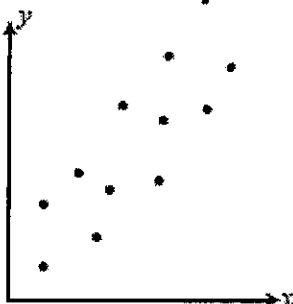
Strong positive linear association

$r > 0$ and very close to 1



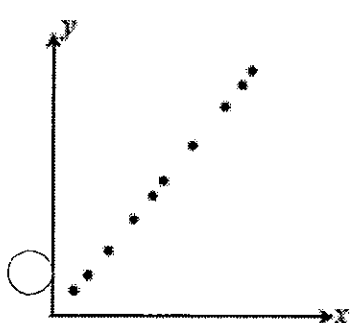
Moderately positive linear association

$r > 0$ and fairly close to 1



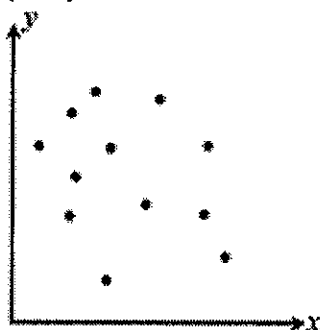
Perfect positive linear association

$r = 1$



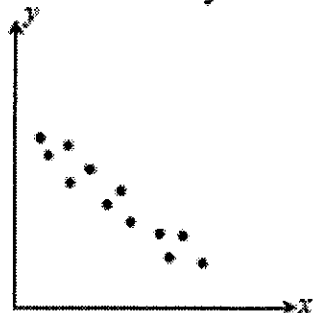
No positive correlation

$r = 0$



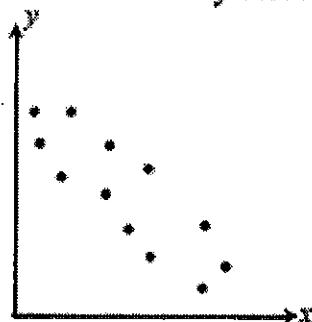
Strong negative linear association

$r < 0$ and very close to -1



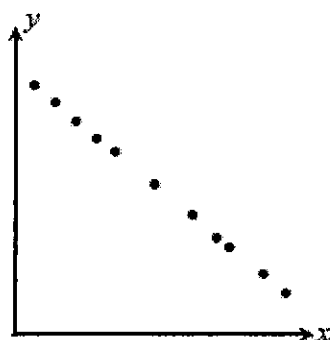
Moderate negative linear association

$r < 0$ and fairly close to -1



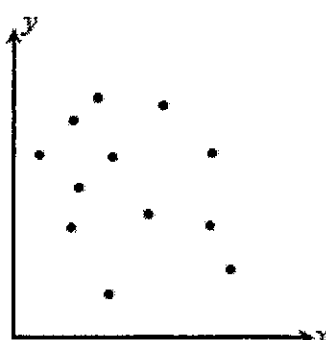
Perfect negative linear association

$$r = -1$$



No negative correlation

$$r = 0$$



The following table provides different examples of r and how to interpret these values.

r	Interpretation
1	Perfect positive association
0,9	Strong positive association
0,5	Moderate positive association
0,2	Weak positive association
0	No association
-0,2	Weak negative association
-0,5	Moderate negative association
-0,9	Strong negative association
-1	Perfect negative association

EXAMPLE 2

The following table shows information about the age of eight fossils of a particular species of ape found in South Africa and the interior volume of their skulls.

Age in millions of years	2,4	2,0	0,8	1,1	1,6	0,7	0,3	1,4
Volume of skull (cm^3)	38	82	174	171	89	159	206	129

Calculate the correlation coefficient (r) and comment on the strength of the linear association.

Solution

A possible calculator programme is as follows:

MODE 2

PRESS 2: A + BX

ENTER DATA POINTS:

IN AGE COLUMN (X)

2.4 = 2.0 = 0.8 = 1.1 = 1.6 = 0.7 = 0.3 = 1.4 =

○ IN COLUMN (Y)

38 = 82 = 174 = 171 = 89 = 159 = 206 = 129 =

THEN PRESS AC

THEN PRESS SHIFT 1

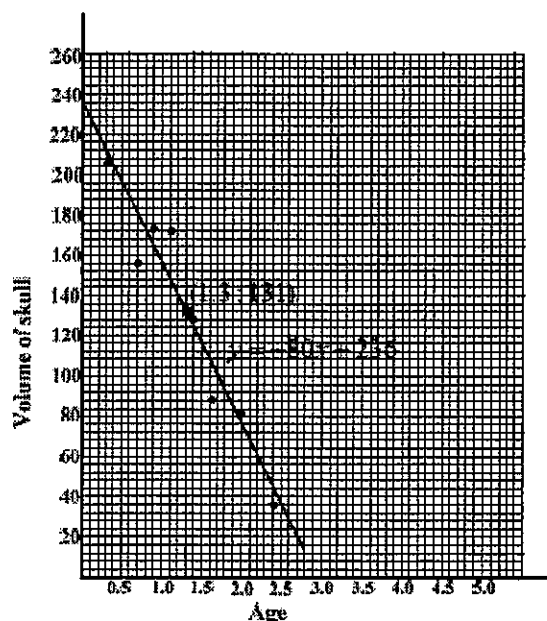
THEN PRESS 5: REG

THEN PRESS 3: $r =$ TO GET THE VALUE OF r WHICH IS $r = -0,9679973645$

THEN PRESS MODE 1 TO GET BACK TO NORMAL MODE

Hence $r = -0,9679973645$

Therefore there is a **strong negative linear association** between the variables.



EXERCISE 2:

1. Paper 3 – 2013

Mr Ryan is a retired teacher who supplements his pension by mowing lawns for customers who live in his neighbourhood.

As part of a review of his charges for this work, he measures the approximate areas (x) (in m^2) of a random sample of 12 of his customers' lawns and notes the time (y) in minutes, that it takes him to mow these lawns.

His results are shown in the table.

Area (x) (m^2)	360	120	845	602	1 190	530	245	486	350	1 005	320	250
Time (y) (minutes)	50	28	130	75	120	95	55	70	48	110	55	60

- (a) Use your calculator to determine the equation of the least squares regression line. Give your answers correct to 4 decimal digits.

(5)

- (b) Calculate the value of r , the correlation coefficient for the data, correct to 4 decimal digits.

(2)

- (c) Given that Mr Ryan charges a flat call out fee of R150, as well as R50 per half hour (or part thereof), estimate the charge for mowing a customer's lawn that has an area of 560 m^2 .
(For example: 100 minutes would be taken as 2 hours)

(3)

- (d) The local high school want Mr Ryan to mow their rugby field which is rectangular, 100 metres long by 70 metres wide. Should you use the regression equation found in (a) to calculate the time it would take to mow this area? Give a reason for your answer,

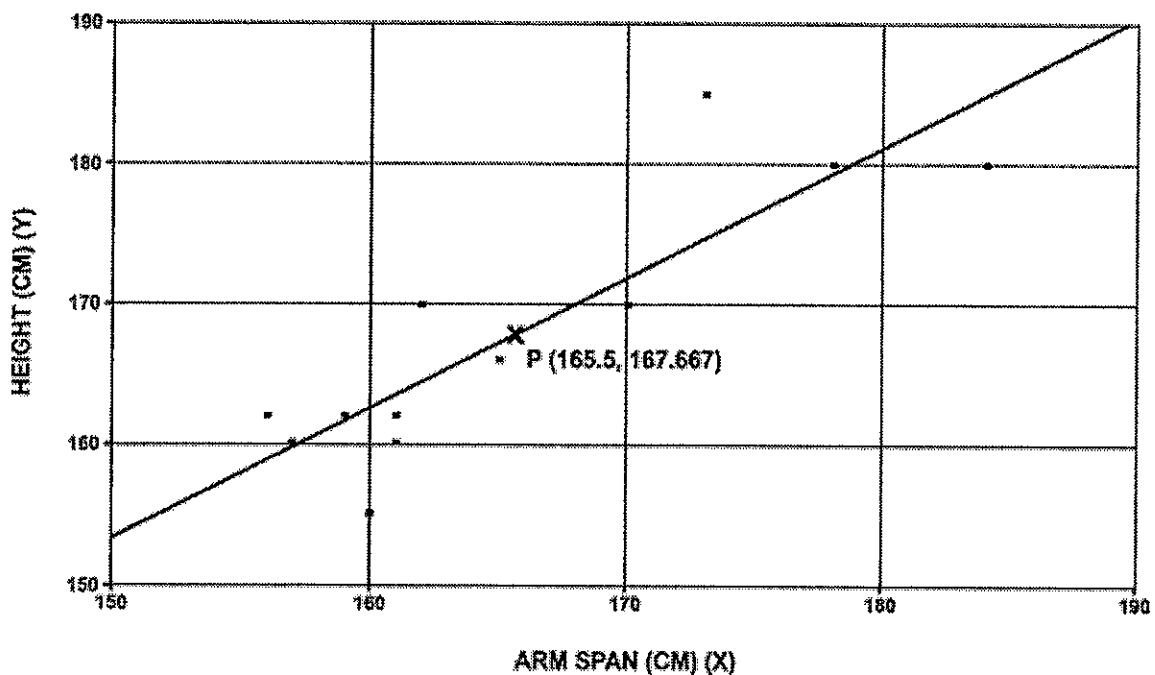
(2)
[12]

2. PAPER 3-2012

Measurements were collected from 12 people to determine whether people with longer arms are taller than people with shorter arms.

Candidate	Arm span (cm) (x)	Height (cm) (y)
A	156	162
B	157	160
C	159	162
D	160	155
E	161	160
F	161	162
G	162	170
H	165	166
I	170	170
J	173	185
K	178	180
L	184	180

The regression line is drawn below.



- (a) Use your calculator to determine the equation of the least squares regression line. Give your answers correct to 4 decimal digits.

_____ (5)

- (b) Marie's arm span is measured to be 160 cm. What would her estimated height be?

_____ (2)

- (c) The point $(\bar{x}, \bar{y})$ with co-ordinates (165,5; 167,667) is labelled P in the scatterplot. Use this information and the scatterplot to identify and write down, (using a letter A, B, C, ... L), a candidate having below average arm-span with above average height. (The letter must be selected from the table at the beginning of the question.)

_____ (1)

- (d) Suppose the correlation (r) between arm span and height is calculated for two cases:

Case 1: Candidate J is included.

Case 2: Candidate J is excluded.

How would the correlations compare? Choose the correct answer from (i) to (iv) below. Write down only your choice.

- (i) The r values would be the same.
(ii) The r value would be closer to 1 for case 1 than for case 2.
(iii) The r value would be closer to 1 for case 2 than for case 1.
(iv) The r value would be closer to -1 for case 1 than for case 2.

Answer: _____ (1)
[9]

3. PAPER 3-2011

In the table below, the number of households, the number of households headed by single parents and the mean family size have been tabulated for 10 areas in Cape Town.

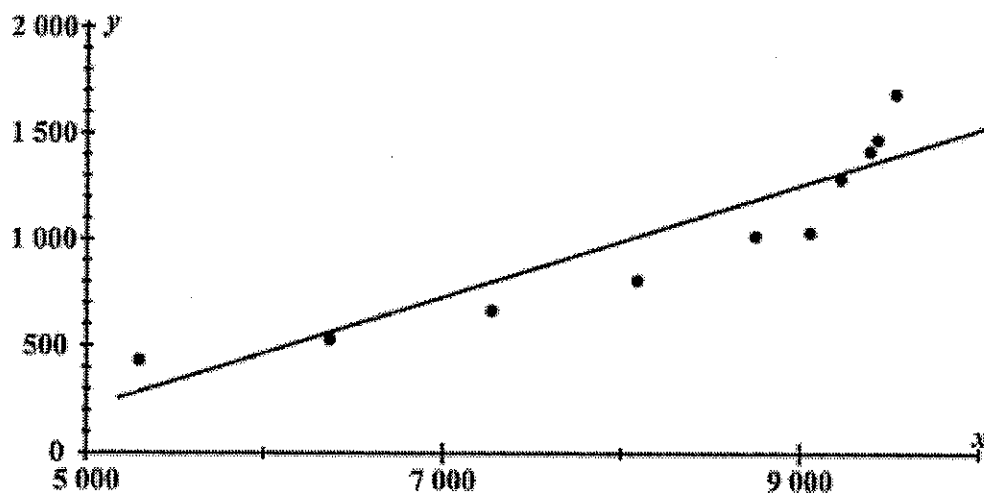
Number of households (x)	Number of households headed by a single parent (y)	Mean family size
5 302	423	3,31
6 359	516	3,14
7 256	657	2,98
8 077	798	2,7
8 729	1 008	2,63
9 028	1 023	2,59
9 205	1 275	2,45
9 379	1 398	2,34
9 412	1 456	2,21
9 517	1 667	2,15

- (a) State whether there is a positive or negative correlation between the number of households headed by single parents and the mean family size.

_____ (1)

- (b) When predicting the number of households headed by single parents (y) on the number of households (x), the equation of the regression line (drawn below) is given by:

$y = A + Bx$ with a correlation coefficient r



- (1) Use your calculator to determine the values of A, B and r . Give answers correct to 4 decimal digits.

A = _____

B = _____

r = _____ (6)

- (2) Use this regression line equation to predict the number of households (to the nearest unit) headed by single parents if there are 15 000 households.

(2)

- (3) Would you have confidence in this prediction for 15 000? Justify your answer.

(2)
[11]

4. PAPER 3-2010

QUESTION 5

A law firm, Justice and Son, monitored the time spent on consultations with a random sample of 120 clients. The times, to the nearest minute, are summarised in the table below:

Time (minutes)	Number of Clients
10 – 14	4
15 – 19	5
20 – 24	18
25 – 29	36
30 – 34	21
35 – 39	25
40 – 44	7
45 – 49	3
50 – 54	1

- (a) Estimate:

- (1) the mean of the times.

(3)

- (2) the percentage of clients who spent less than the mean time in consultation.

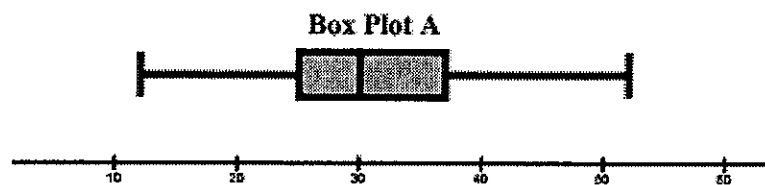
(2)

- (3) the standard deviation of the times.

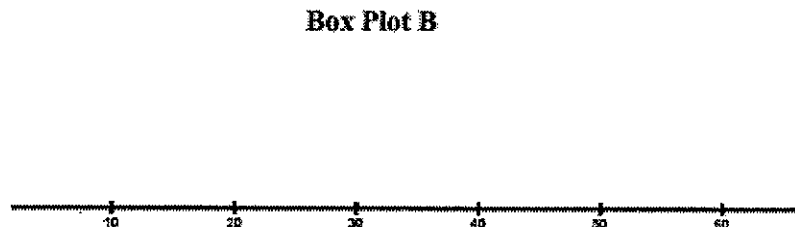
(3)

- (b) A Box and Whisker plot of the data is drawn, in **Box Plot A** below, with the following: Minimum = 12 minutes; $Q_1 = 25$; $Q_2 = 30$; $Q_3 = 37$ and Maximum = 52.

Data regarding consultation times is captured from a second law firm, Peace and Company. They found that the least time spent was 20 minutes, the longest was 40 minutes and the quartiles were 24, 35 and 36 minutes respectively.



- (1) On the scaled graph in **Box Plot B** below, draw the Box Plot to represent the data from Peace and Company.



(2)

- (2) Using the following headings, compare and contrast the two Box Plots.

- (i) Range and inter-quartile range

(2)

- (ii) Skewness

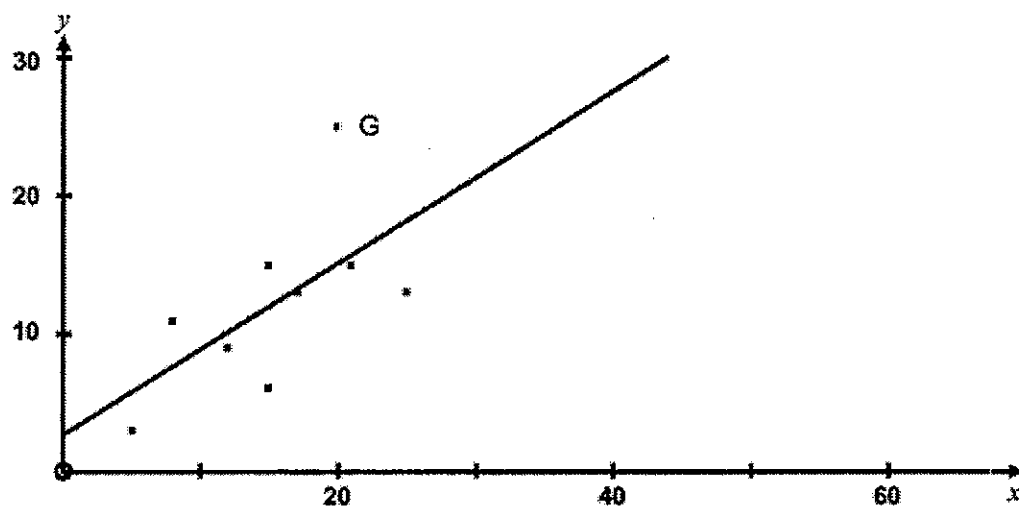
(2)
[14]

QUESTION 7

A supermarket wants to survey how long in seconds (y) it takes to scan (x) items at a till point.

They decide to select the results from 9 shoppers. The results in the table below are plotted in the scatterplot.

	A	B	C	D	E	F	G	H	I
x (no of items)	5	8	12	15	15	17	20	21	25
y (time in seconds)	3	11	9	6	15	13	25	15	13



- (a) Use your calculator to find the equation of the regression line, drawn on the scatterplot, which best fits the given data.

(4)

- (b) Calculate the value of r , the correlation coefficient for the data.

(2)

- (c) Estimate the time that the teller would take to scan 21 items at the till.

(2)

- (d) It was found that an error was made with the time value of point G (20; 25). In fact the point should have been (20; 20).

If the correlation coefficient for the new least squares line of best fit was now calculated, would it indicate a stronger or a weaker correlation? Validate your answer.

(3)

[11]



12